NORTHERN ARIZONA UNIVERSITY

School of Informatics, Computing, and Cyber Systems

CS 212 - Project Phase – 1 Report

Follow the instructions provided in the grey text. Replace the grey text with actual information.

Group Details:

Student 1 Name	Daniel Raygoza
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Student 2 Name	Aaron Skol
Student 2 Email	ars2358@nau.edu
Student 3 Name	Derrick Iron
Student 3 Email	dki4@nau.edu

Phase-1 Details:

Project Title	Course Enrollment Portal
Phase -1 Components	A Weekly Schedule Viewer component. A Course Map component. The Catalog page + dynamic layout. Course List Item. Azure deploy script. Proper JSON dataset for all the courses and course sections. Complete endpoints for: GET /api/courses/:id GET /api/courses?... GET /api/students/:id/courses\ DELETE /api/students/:id/courses/:id/ GET /api/students/:id/schedule POST /api/students/:id/courses/:id/

Task Status:

Avoid incorporating generic information, as it may result in penalties. Copy all the phase-1 tasks listed in the proposed project plan.

Task No	Task Details	Assigned To	Task Status
1	Create a reusable *Weekly Schedule Viewer* component in HTML, and then migrate it into JavaScript by creating a function to insert the HTML anywhere we desire. This component will display the user's current schedule and how adding certain classes will change their schedule.	Daniel	Completed
2	Create a reusable *Course Map* component in HTML, and then migrate it into JavaScript by creating a function to insert the HTML anywhere we desire. This component will show the route the student will have to take for each course, and show the locations of the courses during course search.	Daniel	Completed
3	Create a dynamic layout for the full application that includes the navbar and content for the mobile view, and the navbar, sidebar, and content for the desktop view. Dynamically swapping between them based on screen width.	Daniel	Completed
4	Create a reusable *Course Item List* component to show the list of courses that we fetch from the server. Then, integrate it into a	Daniel	Completed

	Catalog page. The page will have searching and filtering built in.		
5	Write a deployment script to automatically deploy our app to Azure, potentially integrating it with GitHub Actions.	Daniel	Completed
6	Initialize the backend. Set up the folder structure, create a server.js, install dependencies from npm such as nodemon and express, and configure environment variables in .env.	Aaron	Completed
7	Take a large-scale dataset of courses that are offered at NAU from "https://api.extended.nau.edu/api/CourseCatalog"(no longer available, but cached locally in JSON format) and parse it to extract prerequisites, corequisites, and more. Then, for each course, generate an array of class times and waitlist statuses, and finally combine all the data into a single JSON array.	Aaron	Completed
8	Implement GET /api/courses/:id and /api/courses?q=x&limit=y&offset=z. Using url query parameters to filter the list of courses in the catalog. Some filters include: q=search_query limit=pagination_limit offset=pagination_offset instructor=instructor_query subject=subject_id etc. Noting that search_query searches generally through all the information in a given course.	Aaron	Completed

9	Implement GET /api/students/:id/courses. Fetching the students' current enrolled courses based on their student ID. Allowing for a body to the request to contain {additional_course_ids: []} to display.	Aaron	Completed
10	Implement a JSON database to store information about students. Storing their enrolled courses, notifications, and waitlisted courses.	Aaron	Completed
11	Develop all of the boilerplate for the entire backend API, instantiating an Express app, exposing a port, and creating empty API route paths. (GET /api/courses/:id, GET /api/courses?limit=x&offset=y,POST /api/students/:id/course/:id/enroll, DELETE /api/students/:id/courses/:id/, POST /api/students/:id/course/:id/waitlist, DELETE /api/students/:id/courses/:id/waitlist, etc…)	Derrick	Completed
12	Implement POST /api/students/:id/courses/:id/enroll. Checking for conflicts with the current schedule and then properly updating the database to reflect the new open seats. This will also trigger a notification to be added to the student's data.	Derrick	Completed

13	Implement DELETE /api/students/:id/courses/:id/drop. Ensure proper updating of the database to reflect the new open seats. This will also trigger a notification to be added to the student's data.	Derrick	Completed
14	Implement GET /api/students/:id/schedule. Fetching the students' current schedule for their course to be displayed on the *Weekly Schedule Component*.	Derrick	Completed
15	Implement a JSON database to store the override requests from the override request. Storing the information like {id: 0, name: "", email: "", sid: "", course_id: "", reason: "", date: ""}.	Derrick	Completed

New Tasks or Modified Tasks (Optional):

List all newly created tasks or tasks modified from your project proposal. Feel free to add more rows to the following table. **Fill the below section, only if you have any new tasks or modifications in the Phase-1 tasks list or else skip it.**

Task Type	Task Details	Assigned To	Task Status
New/ Modified	Explain the Task in detail (not in general)	Name of the team member	Completed/ Backlog (Pending)/ Modified
New/ Modified	Explain the Task in detail (not in general)	Name of the team member	Completed/ Backlog (Pending)/ Modified

Challenges and Solutions Implementation:

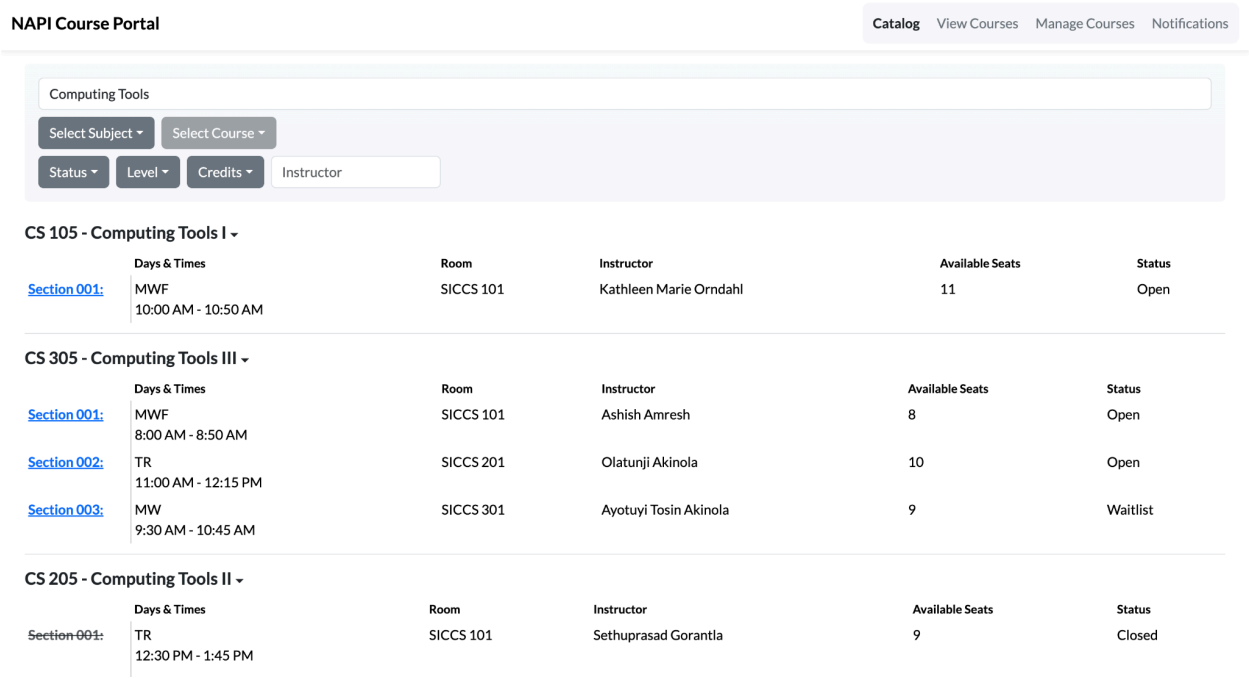
List at least 4 challenges faced by your team during phase-1 development and elaborate on the solutions you have implemented. Feel free to add more rows to the following table.

Details of the Challenge	Solution Implemented
The raw JSON data on the NAU course catalog was not usable in its current state. The values for the requisites and course descriptions were single entry strings that seemed to be read explicitly off the course page. For example, courses that have both pre-requisites and co-requisites would both be written in the same tag producing a single string with both attributes inside. The course times were also formatted similarly, and had to be extracted individually.	We created a python script that was able to parse the JSON data using regex to split the individual entries, and then used list comprehension to merge them back together in order to form lists of lists, then merged them back into a single JSON string to convert the data into a usable format for our webpage.
The routing was initially messy leading to paths, imports, and endpoints being undefined. This was mainly a consequence of us not clearly defining strict file structuring in the repository. Incorrect file pathing was used, and there was once a time where our npm metadata was compromised	We sorted out those issues one by one, essentially refactoring the entire backend. The biggest drawback was the routing becoming very obfuscated, and we went back and cleaned it up by creating modules and helper functions to make the routes way cleaner. This also makes it easier to implement more routes and scale the webpage further.
The original idea to implement the Course Map component was to render a custom map of the NAU campus; however, we realized that would be a much more daunting task than it seemed at the time.	After some research on map-related libraries, we found the best solution to implement this was to use LeafletJs which allows us to use a simple JavaScript interface to render the NAU map and markers using Bootstrap icons.
My task was to add continuous delivery. Due to having to reset my phone, I was unable to access the Microsoft Authenticator codes required to access Azure.	Luckily, IT got to the ticket in time, and I was able to then access Azure, then I created the web app and integrated our project with GitHub actions as stated in the task description.

Phase-1 Developed Screens:

- Incorporate webpage screens corresponding to the project phase - 1.
- Capture all screenshots of your phase-1 delivery (**don't include any code screenshots**) and attach them in this section.

[**Attach all the screenshots of the phase-1 development webpage screens here**]



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NAPI Course Portal

Web Programming

Select Subject

Select Course

Status

Level

Credits

Instructor

CS 504 - Web Programg For Teachers

CS 212 - Web Programming I

CS 312 - Web Programming II

	Days & Times	Instructor	Status
Section 001:	TR 10:00 AM - 11:15 AM	Kathleen Marie Orndahl	Open

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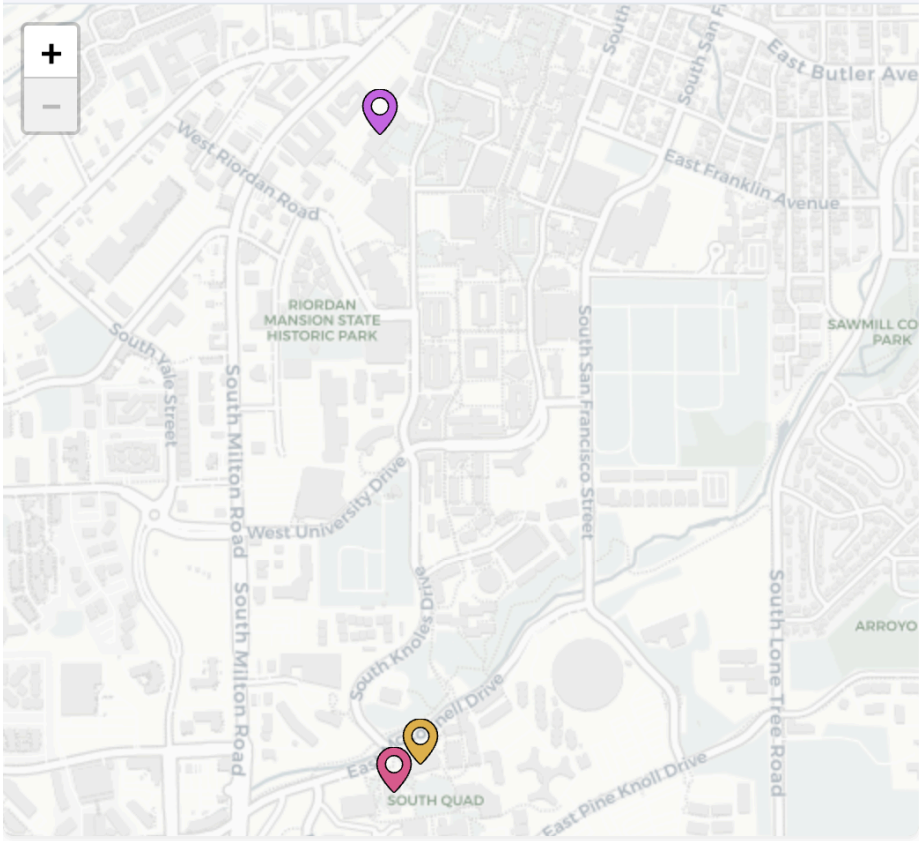
School of Informatics, Computing, and Cyber Systems

Campus Map

All | Mon | Tue | Wed | Thu | Fri